

Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- **Explain** an error (not **fix** your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

```
PS C:\Users\Student\James\IT314AB_Gulfan\my_first_flutter_app> git branch
* feature/data-ui
main
```

🚩 Flag 1 — Identify the Data

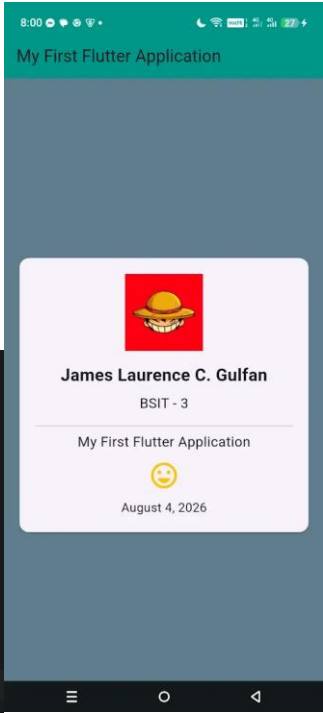
Name	String
Age	Int
Height	double
Status	bool

🚩 Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

Create variables for at least:

```
void main() {  
  runApp(const MyApp());  
}  
  
String name = "James Laurence C. Gulfan";  
String courseSection = "BSIT - 3";  
int age = 19;  
String birthday = "August 20, 2006";  
String hobby = "Guitar";  
double height = 5.5; //  
bool isStudent = true;
```



🚩 Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

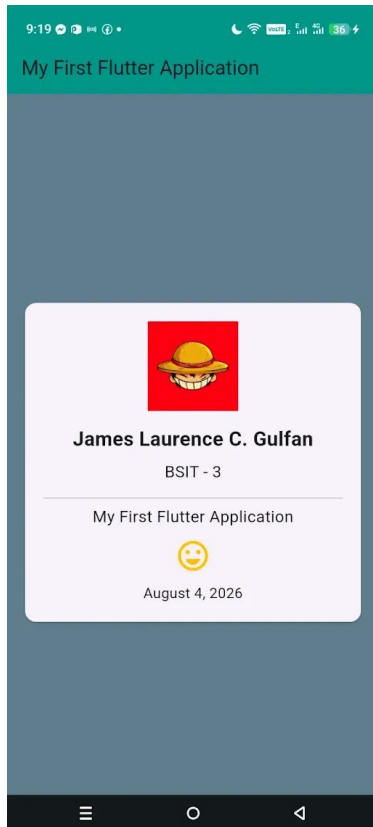
Your profile should now display information using the variables you created.

Test:

Change the value of one of your variables.

```
const SizedBox(height: 15),
Text(
  name,
  style: const TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
),
const SizedBox(height: 10),
Text(courseSection, style: const TextStyle(fontSize: 18)),
const SizedBox(height: 8),
const Divider(),
const Text(
  "My First Flutter Application",
  style: TextStyle(fontSize: 18, fontStyle: FontStyle.italic),
),
const SizedBox(height: 10),
const Icon(
  Icons.sentiment_very_satisfied,
  color: Colors.amber,
  size: 40,
),
Row(
  children: [
    Text("AGE: $age"),
    const SizedBox(width: 20),
    Text("Birthday: $birthday"),
```

```
),
Text(
  "Hobby : $hobby",
  style: const TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
),
const SizedBox(height: 10),
Text("Height: $height ft"),
Text("Student Status: $isStudent"),
],
```



🚩 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

Use the variable when displaying the image.

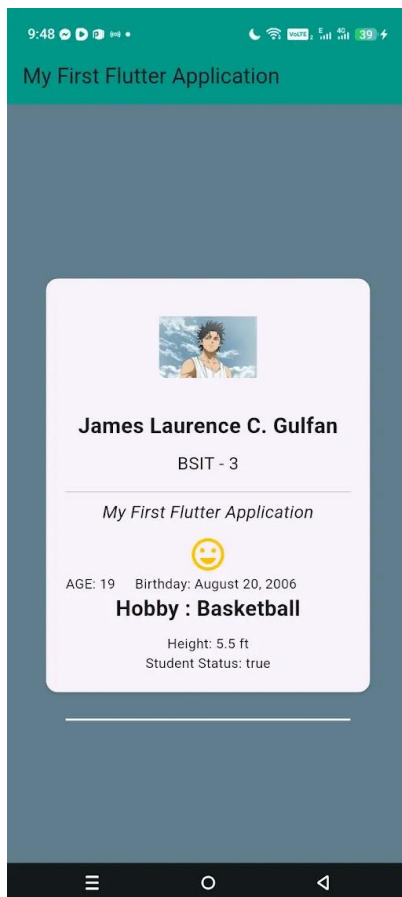
Then change the variable to another registered image.

The image displayed by the application should change.

Screenshot:

- Image variable
- Image widget using the variable
- Updated application

```
String profileImage = 'assets/da.jfif';
String name = "James Laurence C. Gulfan";
String courseSection = "BSIT - 3";
int age = 19;
String birthday = "August 20, 2006";
String hobby = "Basketball";
double height = 5.5;
bool isStudent = true;
```



```
assets:
  - assets/download.jpg
  - assets/da.jfif
```

🚩 Flag 5 — Five Profiles, One Design

Objective: Create at least **five different profile cards** using the same visual design.

Each profile must contain at least:

- Image
- Name
- Course & Section
- Age
- Hobby

The five profiles must contain different information.

Important

All five profiles should use the **same visual design**.

You are demonstrating that the same UI can represent different data.

Screenshot:

- Your 5 different profile data

```
List<Map<String, dynamic>> profiles = [
  {
    'image': 'assets/da.jfif',
    'name': "James Laurence C. Gulfan",
    'courseSection': "BSIT - 3",
    'age': 19,
    'hobby': "Basketball",
  },
  {
    'image': 'assets/john.jfif',
    'name': "John Deacon",
    'courseSection': "BSIT - 3",
    'age': 23,
    'hobby': "Bass",
  },
  {
    'image': 'assets/brian.jfif',
    'name': "Brian May",
    'courseSection': "BSIT - 3",
    'age': 24,
    'hobby': "Sleeping",
  },
]
```

```
{
  'image': 'assets/fred.jfif',
  'name': "Fred Merc",
  'courseSection': "BSIT - 3",
  'age': 20,
  'hobby': "Singing",
},
{
  'image': 'assets/roger.jfif',
  'name': "Roger Taylor",
  'courseSection': "BSIT - 3",
  'age': 26,
  'hobby': "Dancing",
},
]
```

Flag 6 — THE HIDDEN PROBLEM

You didn't see your UI?

Objective: Make all five profiles accessible to the user.

You just added five profile cards.

Now run your application.

Something is wrong.

Look carefully at your emulator.

Can you see all five profiles?

Maybe you can see the first one...

Maybe part of the second...

But where are the rest?

That's right.

You can't see them.

Your UI is larger than the available screen.

The profiles exist.

The data exists.

The widgets exist.

But the user can't access everything.

Your Mission:

Figure out how Flutter allows users to **scroll through content that is larger than the available screen**.

Research Flutter's **scrolling widgets** and determine which one is appropriate for your current layout.

You must make it possible to scroll through all five profiles.

Requirement:

The user must be able to:

Scroll → see Profile 1 → Profile 2 → Profile 3 → Profile 4 → Profile 5

without changing the size of the screen.

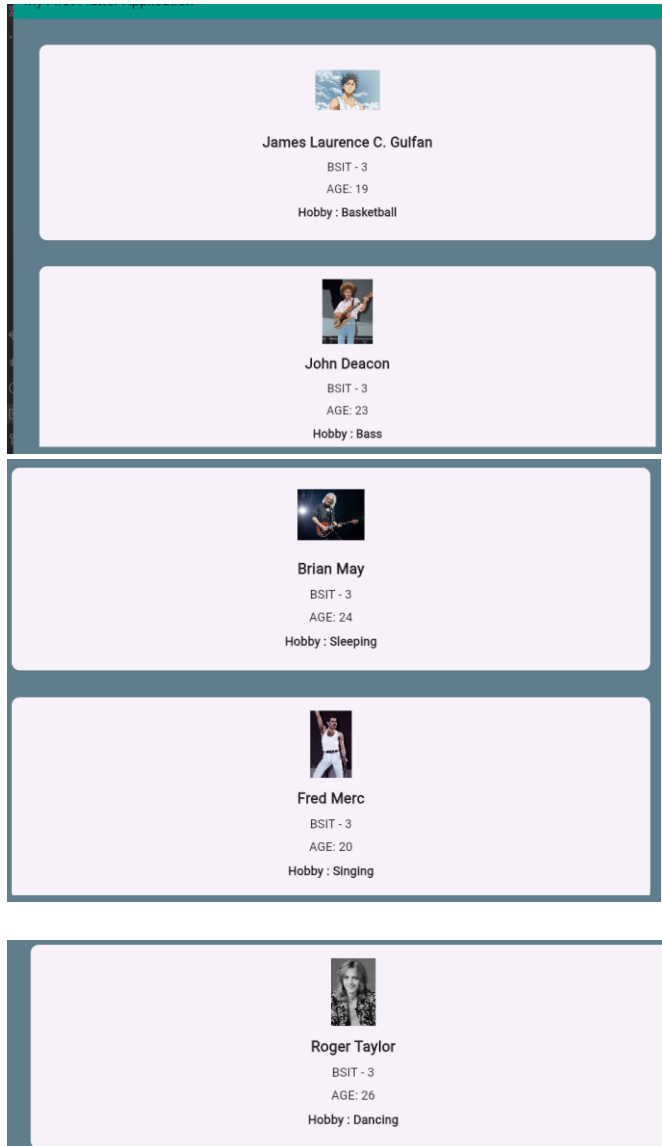
Screenshot:

- The scrolling widget in your code
- Top portion of the application
- The rest of the portion of the application showing the later profiles

PS: maam nag chrome kay naguba akong cellphone in the middle sa activity.

```
child: ListView(  
  children: [  
    ...profiles.map(  
      (profile) => Card(  

```



🚩 Flag 7 — Survive the Database

Objective: Prepare your UI for incomplete data.

Now imagine your profiles aren't manually created.

Someday, the data will come from a database.

And real data isn't always perfect.

Some information may be:

- Missing
- Empty
- `null`
- Not provided

Create at least **three profiles with missing information**.

For example:

Profile A

Name: Maria

Course: BSIT

Hobby: **missing**

Profile B

Name: Juan

Course: **missing**

Hobby: Gaming

Profile C

Name: **missing**

Course: MMA

Hobby: Drawing

Your application must **not crash** when information is missing.

Instead, provide an appropriate fallback.

Examples:

Hobby: Not provided

Course: Unknown

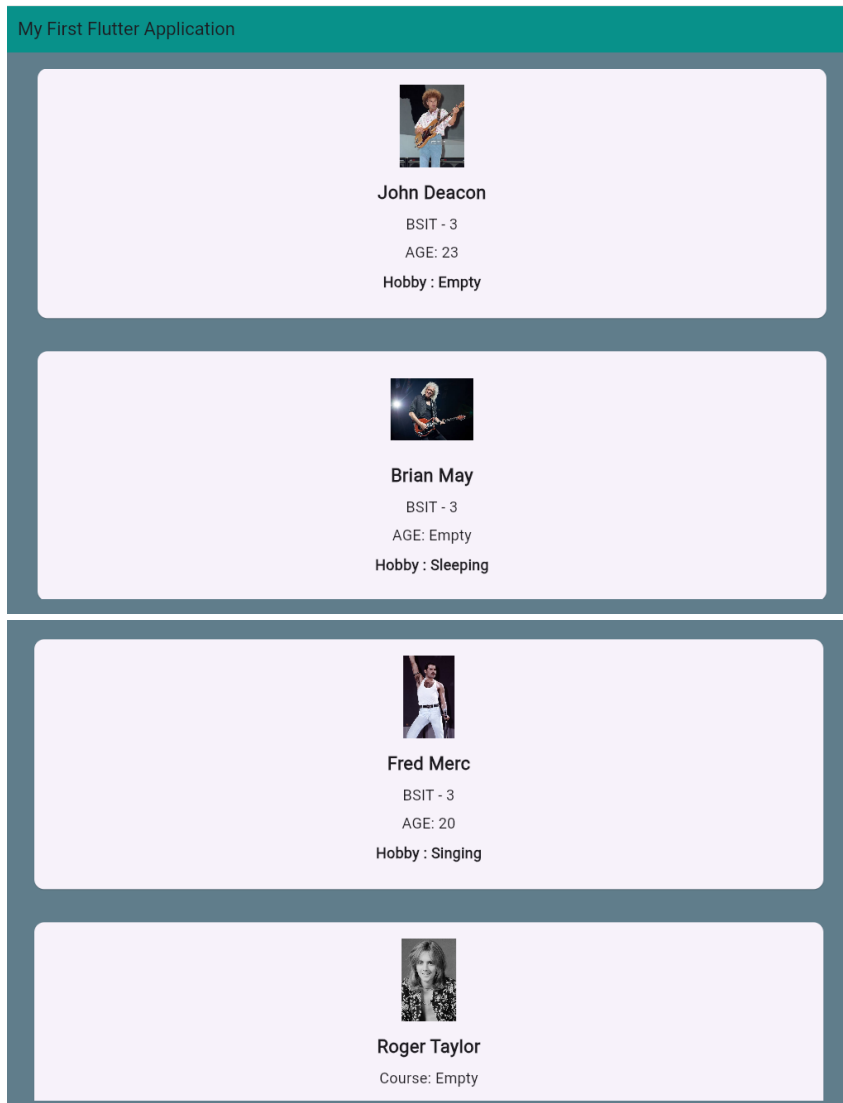
Challenge

Your placeholders should look intentional and clean.

Screenshot:

- Incomplete profile data
- Application displaying fallback values
- Application still functioning correctly

```
Text(  
  profile['name'] ?? "Name: Empty",  
  textAlign: TextAlign.center,  
  style: const TextStyle(  
    fontSize: 22,  
    fontWeight: FontWeight.bold,  
  ),  
)  
,  
const SizedBox(height: 10),  
Text(  
  profile['courseSection'] ?? "Course: Empty",  
  textAlign: TextAlign.center,  
  style: const TextStyle(  
    fontSize: 18,  
  ),  
)  
,  
const SizedBox(height: 8),  
Text(  
  "AGE: ${profile['age'] ?? "Empty"}",  
  textAlign: TextAlign.center,  
  style: const TextStyle(  
    fontSize: 18,  
  ),  
)  
,  
const SizedBox(height: 10),  
Text(  
  "Hobby : ${profile['hobby'] ?? "Empty"}",  
  textAlign: TextAlign.center,  
  style: const TextStyle(  
    fontSize: 18,  
    fontWeight: FontWeight.bold,  
  ),  
)
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/data-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
PS C:\Users\Student\IT314AB_Gulfan\my_first_flutter_app> git status
On branch feature/data-ui
Your branch is up to date with 'origin/feature/data-ui'.
```

```
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   assets/brian.jfif
    new file:   assets/da.jfif
    new file:   assets/fred.jfif
    new file:   assets/john.jfif
    new file:   assets/roger.jfif
    modified:   lib/main.dart
    modified:   pubspec.yaml
```

```
PS C:\Users\Student\IT314AB_Gulfan\my_first_flutter_app> git branch
  Feature/Widgets
* feature/data-ui
  main
```

```
PS C:\Users\Student\IT314AB_Gulfan\my_first_flutter_app> git commit -m "feat: submit capture the flag 3"
[feature/data-ui 746c052] feat: submit capture the flag 3
 7 files changed, 104 insertions(+), 47 deletions(-)
 create mode 100644 my_first_flutter_app/assets/brian.jfif
 create mode 100644 my_first_flutter_app/assets/da.jfif
 create mode 100644 my_first_flutter_app/assets/fred.jfif
 create mode 100644 my_first_flutter_app/assets/john.jfif
 create mode 100644 my_first_flutter_app/assets/roger.jfif
```

```
PS C:\Users\Student\IT314AB_Gulfan\my_first_flutter_app> git push origin feature/data-ui
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 118.38 KiB | 23.68 MiB/s, done.
Total 12 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/james-wreck/IT314AB_Gulfan.git
   15e0c84..746c052  feature/data-ui -> feature/data-ui
```

```
PS C:\Users\Student\IT314AB_Gulfan\my_first_flutter_app> git add .
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
```